Academic year 2024-2025 (Odd Semester)

CIE I: Scheme and Solution

Course	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING		
Date	Dec 2025	Dec 2025	Dec 2025
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Sem	V	V	

SL No.	Part A	M
1.1	A heuristic function $h(n)$ estimates the cost of the cheapest path from a node n to the goal. It is used in informed search algorithms (like A*) to guide the search efficiently by prioritizing nodes that appear closer to the goal.	1
1.2	A node's value becomes worse than an already explored alternative, so further exploration cannot affect the final decision.	1
1.3	Choose the label at random OR choose the class with higher global frequency, Assign higher weight to the closer neighbor—if equal distance, use another distance metric or use k > 1 to avoid ties.	2
1.4	All features are assumed to be independent of each other *given the class label*.	2
1.5	Explanation with respect to Random Forests reduce correlation among trees and therefore outperform Bagging.	2
1.6	MAX → maximize value, MIN → minimize value, They alternate levels in the game tree, working to optimize opposing objectives.	2

	Part B	
1.a	A* finds a shortest path from a start node `S` to a goal `G` by combining: • `g(n)` — the exact cost from the start `S` to node `n` (so far), • `h(n)` — a heuristic estimate of the cost from `n` to the goal, • `f(n) = g(n) + h(n)` — the priority value A* uses to decide which node to expand next ..3M Suitable example..2M	
1.b	• Left MIN returns min(4,8,2) = **2** • α for MAX = 2 • Right MIN sees 5 → β = 5 • Next value = 1 → β = 1	5

	• MAX compares 2 vs 1 → takes **2** • No pruning occurs because all children must be checked.	

2.a	Compute class priors: Count Buy = Yes and No:…2M
	• Yes: records {3,4,5,7,9} → 5 examples
	• No: records {1,2,6,8,10} → 5 examples
	P(Yes)=5/10=0.5,P(No)=5/10=0.5
	Compute categorical conditional probabilities…2M
	Age Group = Middle
	• Among Yes: Middle occurs in records 3 and 5 → 2/5 = **0.40**
	• Among No: Middle occurs in record 8 → 1/5 = **0.20**
	Marital Status = Divorced
	• Among Yes: Divorced occurs in records 5 and 9 → 2/5 = **0.40**
	• Among No: Divorced occurs **0 times**. Apply Laplace smoothing (k = number of categories = 3: Single, Married, Divorced):
	$$P(\text{Divorced} \mid \text{No}) = \frac{0+1}{5+3} = \frac{1}{8} = 0.125$$
	Compute Gaussian PDF…4M
	Split spending values by class:
	• Spending | Yes: 80, 90, 65, 95, 85
	  • Sum = 80+90+65+95+85 = 415
	  • Mean (μ_Yes) = 415 / 5 = **83.0**
	  • Variance (population) = [(80−83)² + (90−83)² + (65−83)² + (95−83)² + (85−83)²] / 5 = [9 + 49 + 324 + 144 + 4] / 5 = 530 / 5 = **106.0**
	  • Std dev σ_Yes = √106 ≈ **10.296**
	• Spending | No: 55, 70, 60, 50, 45
	  • Sum = 280 → Mean (μ_No) = 280 / 5 = **56.0**
	  • Variance = [1 + 196 + 16 + 36 + 121] / 5 = 370 / 5 = **74.0**
	  • Std dev σ_No = √74 ≈ **8.602**
	Gaussian PDF:
	$$p(x \mid C) = \frac{1}{\sqrt{2\pi}\,\sigma_C} \exp\Big(-\frac{(x-\mu_C)^2}{2\sigma_C^2}\Big)$$
	For Yes:
	• (x − μ_Yes) = 75 − 83 = −8 → $(x-\mu)^2$ = 64
	• Exponent = −64 / (2×106) = −64/212 ≈ −0.301887
	• exp(exponent) ≈ e^{−0.301887} ≈ **0.7395**
	• Denominator = √(2π) × σ_Yes ≈ 2.50663 × 10.296 ≈ **25.80**
	• So
	$$P(75 \mid \text{Yes}) \approx \frac{0.7395}{25.80} \approx **0.02866**$$
	For No:
	• (x − μ_No) = 75 − 56 = 19 → squared = 361
	• Exponent = −361 / (2×74) = −361/148 ≈ −2.43919
	• exp(exponent) ≈ e^{−2.43919} ≈ **0.0874**
	• Denominator = 2.50663 × 8.602 ≈ **21.55**
	• So
	$$P(75 \mid \text{No}) \approx \frac{0.0874}{21.55} \approx **0.00406**$$
	Compute class-conditional joint likelihood:..1M
	For Yes:
	$$P(X \mid \text{Yes}) = P(\text{Age=Middle} \mid \text{Yes}) \cdot P(\text{Divorced} \mid \text{Yes}) \cdot P(75 \mid \text{Yes})$$
	$$= 0.40 \times 0.40 \times 0.02866 = 0.16 \times 0.02866 \approx **0.004586**$$
	For No:
	$$P(X \mid \text{No}) = P(\text{Middle} \mid \text{No}) \cdot P(\text{Divorced} \mid \text{No}) \cdot P(75 \mid \text{No})$$
	$$= 0.20 \times 0.125 \times 0.00406 = 0.025 \times 0.00406 \approx **0.0001015**$$
	Final Prediction… Buy=Yes -1M
		10

3.a	**Logistic Regression** is a **supervised machine learning algorithm** used mainly for **binary classification**, although extensions exist for multi-class classification (e.g., multinomial logistic regression)…1M Explanation of linear model, sigmoid function, decision rule…1M Working—1M, Limitations- 1M	5
3.b		5

Criterion	Bagging (Bootstrap Aggregation)	Boosting (e.g., AdaBoost, Gradient Boosting)	Random Forests
Type of base learner	Typically **high-variance, unstable** learners (usually **decision trees**).	Weak learners, usually **shallow decision trees (stumps)**.	Same as bagging: **full decision trees** are common.
Method of sample generation	Creates many bootstrap samples (sampling **with replacement** from training set).	Sequential sampling: each new learner focuses on **previously misclassified** samples (weighted sampling).	Bootstrap samples + **random feature selection** at each split.
Bias–Variance trade-off	**Reduces variance**; bias remains roughly the same.	**Reduces bias** significantly; variance can increase.	Strong **variance reduction** + some decrease in bias due to decorrelation of trees.
Robustness to noise/outliers	**High robustness**: averaging reduces noise impact.	**Low robustness**: boosting often overfocuses on noisy or mislabeled points.	**Very robust**: similar to bagging but even better due to feature randomness.
Tendency toward overfitting	Low tendency; stable with many trees.		

4.a	i. Process of Creating many training sets by sampling with replacement from the original dataset; each bootstrap sample has the same size as the original but may contain duplicates and will typically omit some examples. A decision stump is a one-level decision tree: it picks one feature and one threshold (split) and predicts using that single split (e.g., predict +1 if $x < t$, else −1)…..2M ii. Bagging trains many such learners on independent bootstrap samples and aggregates their predictions (majority vote for classification, average for regression). Aggregation averages out the random fluctuations of individual learners, reducing variance…..2M iii. Majority voting (or averaging) eliminates the effect of the few stumps that were misled by noise: correctly trained stumps outvote the noisy ones, restoring a correct ensemble decision…..2M iv.	10

Feature	Single Decision Stump	Bagging (with Stumps)

Accuracy	Low (very simple model)	Higher (ensemble averaging improves results)
Variance	Moderate; sensitive to small changes	Very low due to aggregation of many models
Robustness to Noise	Poor; noisy points can shift threshold	Strong; bootstrapping reduces noise influence
Stability	Unstable near decision boundary	Highly stable due to majority voting

2M

v. Stump 1 → −1, Stump 2 → −1…1M

vi. Final Prediction -1 …1M

5a (10)

i. Initialize weights…1M

Total 5 samples → equal weights: w1=w2=w3=w4=w5=1/ 5=0.2

ii. Try stump thresholds k = 3, 4, 5 → choose stump with lowest error…2M

For k=3, 1 misclassification, For k=4, 1 misclassification, For k=5, 1 Misclassification

iii. Compute weighted error …2M

Weighted error in all case is 0.2.

iv. Compute stump weight α. …2M

$$\alpha_1 = \frac{1}{2}\ln\left(\frac{1-\epsilon}{\epsilon}\right) = \frac{1}{2}\ln\left(\frac{0.8}{0.2}\right) = \frac{1}{2}\ln(4) = \frac{1}{2} \times 1.386 = 0.693$$

v. Update sample weights….2M

For correctly classified samples (1–4)

$$w_i' = 0.2 \cdot e^{-\alpha_1} = 0.2 \cdot 0.5 = 0.1$$

For misclassified sample (5)

$$w_5' = 0.2 \cdot e^{+\alpha_1} = 0.2 \cdot 2 = 0.4$$

vi. Normalize new weights ..1M

$$0.1 + 0.1 + 0.1 + 0.1 + 0.4 = 0.8$$